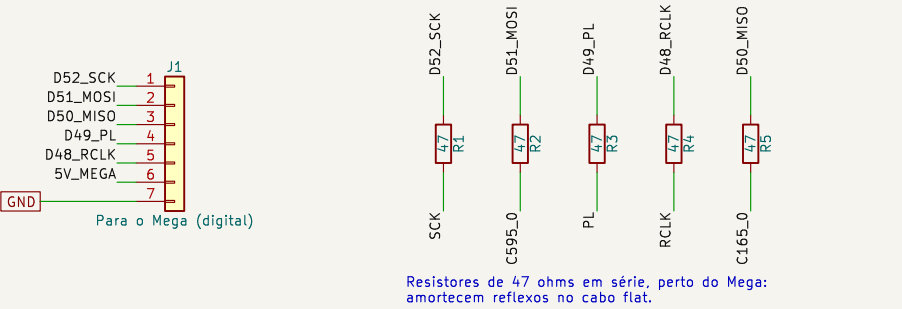
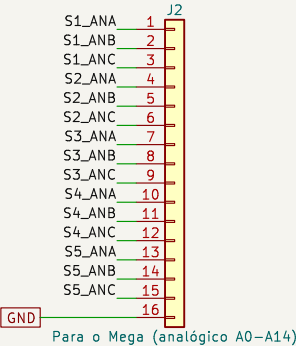


Backplane: liga os módulos ao Mega

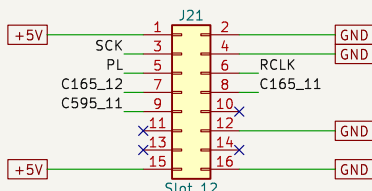
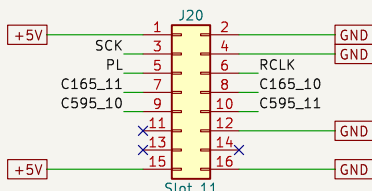
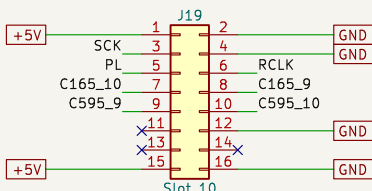
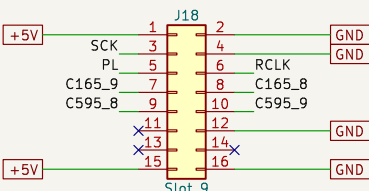
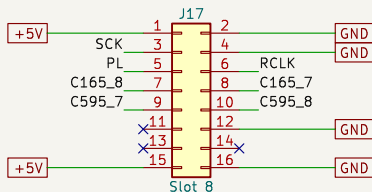
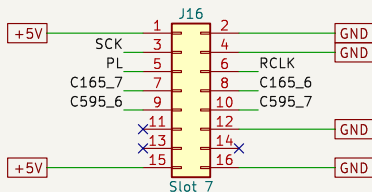
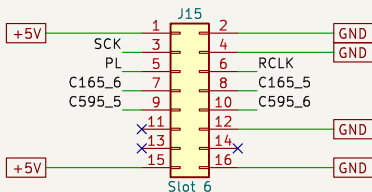
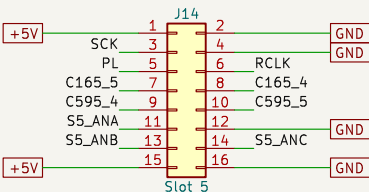
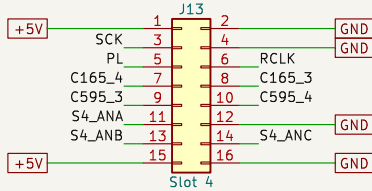
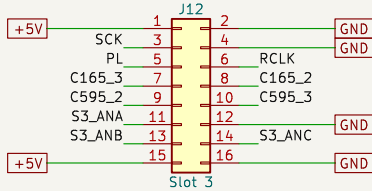
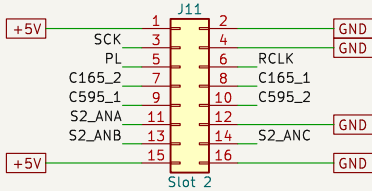
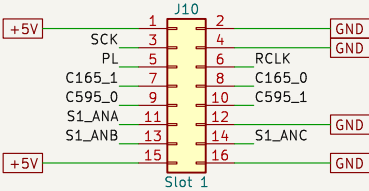
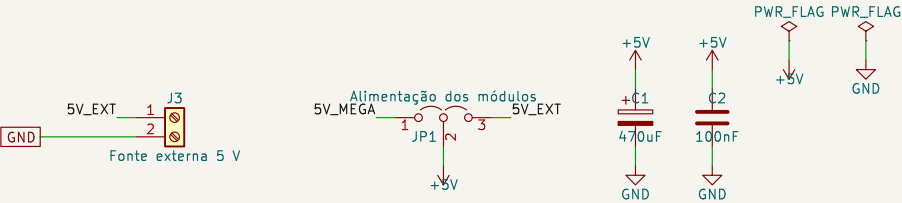
Cada módulo liga num slot por um cabo flat de 16 vias. As entradas de todos os módulos formam uma cadeia de 74HC165 que termina no MISO do Mega; as saídas, uma cadeia de 74HC595 que começa no MOSI.



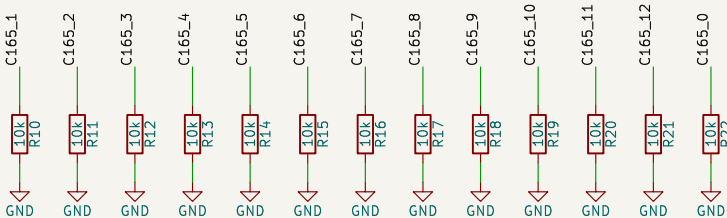
Resistores de 47 ohms em série, perto do Mega: amortecem reflexos no cabo flat.



Para o Mega (analógico A0–A14)



Pull-downs: slot vazio lê 0 em vez de ruído



- Regras
- Módulos em slots seguidos, a partir do slot 1, sem buracos: a cadeia para no primeiro slot vazio.
 - Módulos com joystick ou acelerador vão nos slots 1 a 5, os únicos com linhas analógicas (slot k: A(3k–3) a A(3k–1)).
 - JP1 escolhe de onde vem o +5V dos módulos:
1–2 = 5V do Mega (poucos módulos, alimentado pela USB);
2–3 = fonte externa em J3 (painel completo).
 - O GND do Mega (J1 pino 7) e o da fonte ficam em comum. A ordem dos módulos não importa: cada um se identifica pela etiqueta (chave DIP), e a fila termina onde o Mega lê 0.

Cadeia SPI: 74HC165 para as entradas, 74HC595 para as saídas
12 slots IDC 2x8; slots 1 a 5 com linhas analógicas

KSP Cockpit

Sheet: /
File: backplane.kicad_sch

Title: Backplane: liga os módulos ao Mega

Size: A3
KiCad E.D.A. kicad-
cli 7.0.11+dfsg-1build4

Date: 2026-09-26
Rev: 1
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